

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1621

Oregon City 100 kV
October 27, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1621. The Interconnection Customer proposes a 210.6 MW solar photovoltaic facility interconnecting at the Oregon City 100 kV substation in the Illinois area. The request was received on December 14, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 613 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$72,796,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Decatur Route 51 - Cedar 100 kV	Oxbow Substation - N Bonneville 100 kV L	333.9	136	14.4
Asbury - Decatur Route 51 100 kV	North Bonneville Substation - Oxbow 100 kV L	332.0	134	14.4
Asbury - Brandy Branch 100 kV Line	North Bonneville - Oxbow 100 kV	331.8	131	14.4
Leota - Brandy Branch 100 kV	Oxbow - N Bonneville 100 kV	330.5	128	14.4
Flambeau Hydroelectric Sta 100/1kV Transformer Bank #1	Oxbow - North Bonneville 100 kV	299.6	240	35.1
Flambeau Hydroelectric Sta - Cedar 100 kV	Oxbow Substation - N Bonneville Substation 100 kV	278.2	170	14.4
Flambeau Hydroelectric Sta - Shoreline 100 kV	Oxbow - N Bonneville 100 kV L	271.4	520	60.2
Anza 230-100 kV Autotransformer	Front Range Substation - Shoreline 100 kV	270.4	135	34.9
Shoreline - Front Range 100 kV	N Bonneville - Oxbow 100 kV	265.2	539	60.2
Front Range - Socrates 19 100 kV	North Bonneville Substation - Oxbow 100 kV	265.0	539	60.2
Shawboro - Tocol 100 kV	Shoreline - Front Range 100 kV	255.4	207	62.4
Tocol - Oregon City 100 kV	Front Range - Shoreline 100 kV line	252.5	211	62.4
Flambeau Hydroelectric Sta 230/1 kV Transformer	Oxbow - North Bonneville 100 kV	241.7	290	35.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Davis Crk - Flambeau Hydroelectric Sta 100 kV	Flambeau Hydroelectric Sta 100/1kV Autotransformer	227.2	103	19.8
Shawboro - Sky Vly 100 kV	Shreline - Front Range 100 kV Line	219.8	119	32.0
Price Hill - Davis Crk 100 kV	Flambeau Hydroelectric Station 100/1 kV Autotransformer T1	202.2	125	19.8
Leota - Hoosier Fairview 100 kV	Oxbow Substation - N Bonneville 100 kV L	194.1	131	10.4
Blythe - Oxbow 100 kV	Front Range - Shoreline Substation 100 kV	180.9	673	37.5
N Bonneville - Oxbow 100 kV	Shoreline - Front Range 100 kV line	179.3	686	37.5
Anza - Sky Vly 100 kV	Shoreline - Front Range 100 kV	161.3	142	23.5
Pines Road - Leota Substation 100 kV	Flambeau Hydroelectric Station 100/1 kV Autotransformer Bank 1	158.6	128	7.8
Talentino Sw Sta - Amrad 100 kV	Front Range - Shoreline 100 kV	158.6	130	12.7
Moore Rd - Amrad 100 kV L	Shoreline - Front Range 100 kV	156.1	128	12.7
Enid - Eas Frankfort 100 kV	Flambeau Hydroelectric Station 100 kV/1 kV Auto #1	155.9	108	7.8
Enid - Pines Rd 100 kV	Flambeau Hydroelectric Sta 100/1 kV Auto #1	155.5	121	7.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Decatur Route 51 - Cedar 100 kV	Rebuild line with higher-capacity conductor	\$2,306,000	43% (\$983,000)
Asbury - Decatur Route 51 100 kV	Rebuild line with higher-capacity conductor	\$2,138,000	43% (\$911,000)
Asbury - Brandy Branch 100 kV Line	Rebuild line with higher-capacity conductor	\$2,715,000	43% (\$1,157,000)
Brandy Branch - Leota 100 kV	Rebuild line with higher-capacity conductor	\$3,633,000	43% (\$1,548,000)
Flambeau Hydroelectric Sta 100/1kV Transformer Bank #1	Replace transformer with higher-rated unit	\$3,670,000	40% (\$1,463,000)
Cedar - Flambeau Hydroelectric Sta 100 kV	Rebuild line with higher-capacity conductor	\$481,000	43% (\$205,000)
Shoreline - Flambeau Hydroelectric Sta Substation 100 kV L	Rebuild line with higher-capacity conductor	\$646,000	47% (\$303,000)
Anza 230/100 kV Autotransformer	Replace transformer with higher-rated unit	\$12,715,000	39% (\$4,966,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shoreline - Front Range 100 kV	Rebuild line with higher-capacity conductor	\$1,400,000	47% (\$657,000)
Front Range - Socrates 19 100 kV	Rebuild line with higher-capacity conductor	\$1,486,000	47% (\$698,000)
Tocoi - Sawboro 100 kV	Rebuild line with higher-capacity conductor	\$6,053,000	61% (\$3,697,000)
Tocoi - Oregon City 100 kV	Rebuild line with higher-capacity conductor	\$5,942,000	61% (\$3,629,000)
Flambeau Hydroelectric Sta 230/1kV Transformer Bank	Replace transformer with higher-rated unit	\$12,740,000	40% (\$5,079,000)
Davis Crk - Flambeau Hydroelectric Sta 100 kV	Rebuild line with higher-capacity conductor	\$1,003,000	48% (\$478,000)
Shawboro - Sky Valley 100 kV	Rebuild line with higher-capacity conductor	\$37,128,000	58% (\$21,531,000)
Price Hill - Davis Crk 100 kV	Rebuild line with higher-capacity conductor	\$4,646,000	48% (\$2,214,000)
Leota - Hoosier Fairview 100 kV	Rebuild line with higher-capacity conductor	\$2,534,000	43% (\$1,083,000)
Blythe - Oxbow 100 kV	Rebuild line with higher-capacity conductor	\$1,208,000	46% (\$555,000)
N Bonneville - Oxbow 100 kV	Rebuild line with higher-capacity conductor	\$17,013,000	46% (\$7,824,000)
Anza - Sky Valley 100 kV L	Rebuild line with higher-capacity conductor	\$9,958,000	36% (\$3,617,000)
Pines Rd - Leota 100 kV	Rebuild line with higher-capacity conductor	\$6,974,000	54% (\$3,793,000)
Talentino Sw Sta - Amrad 100 kV	Rebuild line with higher-capacity conductor	\$8,693,000	30% (\$2,590,000)
Moore Rd - Amrad 100 kV L	Rebuild line with higher-capacity conductor	\$5,621,000	30% (\$1,675,000)
Enid - Eas Frankfort 100 kV	Rebuild line with higher-capacity conductor	\$1,803,000	54% (\$981,000)
Enid - Pines Road 100 kV L	Rebuild line with higher-capacity conductor	\$2,132,000	54% (\$1,159,000)
	Total allocated to project		\$72,796,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.